LAPTOP SALES AND PURCHASE SYSTEM
(SDD)

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Session:	**2023-2025**

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BSCS/National College Hasilpur


بسم الله الرحمن الرحيم


اسلامیہ یونیورسٹی بہاولپور

<Laptop Sales and purchase System>

[SOFTWARE DESIGN DOCUMENT]

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[SESSION: 2023-2025| BSCS/NATIONAL COLLEGE HASILPUR

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Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
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CHAPTER NO 1:

INTRODUCTION

1. Introduction

‘’Laptop Sales and Purchase System’’ is used to keep record of whatever the sale from shop. It is also used keep track of remaining balance. It is the project of selling and purchasing the laptop by using this software the shop manager can easily manage their whole system. It provide the faculty of data update, data recovery, data manipulation, data modification and integration. It also provides data of security from unauthorized access**. ‘’Laptop Sales and Purchase System’’** also saves times like in traditional data system. Data recovery and backup is also a major benefit of this system in case of any loss of data. This SRS is going to define the different functions, interfaces and performance of the system. This document is a Desktop- based that is used for the Laptop management system

1.1 Purpose

Main purpose of laptop management system is to better performance of system so that every person an easily interact and perform their actions. Information about the Laptops and their records are easily kept in this system. Laptops are fully functional that consume less power. Actually this system is only made to access all the information. All records are permanently stored in it .Time saving and data correctness is the main benefit of this system. We can have the record of sale and purchase system, so we can deal the customers easily. The main purpose of sale and purchase is the time and place where a retail transaction is completed. We usually preferred our laptops for gaming purposes.

Stock lists and its copy

- Manage costs (internal and external)
- Sales of Laptop on Finance fully supported
- A high quality sales printed
- Unlimited laptop photo graph
- Complete profit and loss analysis on every laptop sold.

1.2Scope

- Best Seller(The bestseller can sell the laptops in best price)
- Bundling of products(The customer can purchased the big box electronics with it’s economic of scope)

- Social media marketing(Through laptops we can do the social media marketing)
- Strategy for information (We can have the purchase price , which also requires monthly payment through a subscription)

1.3 Overview

This document will describe the design of various aspects of the project ‘’Laptop Sales and Purchase System’’ including architectural of project, software components, interfaces, and database design

1.4 Reference Material

In this subsection:

- www.google.com.
- www.w3school.com.
- www.draz.pk.
- www.wikipedia.com
- www.Olx.com

1.5 Definitions and Acronyms

- **Manager** (Manager is a person which has complete authorization. Manager has authority to update stocks or update prices and delete customer record from database etc.)
- **Buyer** (that person who buy product from your shop/mall)
- **Admin** (Login for attraction with application).
- **Modify** (For modification in database and user records).
- **Logout** (For un-attraction with application)
- **RESUME** (To restart something).
- **Framework (**It is a platform for developing application. It is launched by Microsoft. It gives features for developing applications**)**
- **Visual studio (**It is an IDE that support multiple languages like c#, java, .net)
- **DFD (** stands for data flow diagrams)
- **GUI (** stands for graphical user interface)
- **ER Diagrams (**stands for entity relationship diagrams)

CHAPTER NO 2:

SYSTEM OVERVIEW

2.1 Technologies Used

The system is coded with c# by using VISUAL STUDIO Tool. We will use local/internal database. The System is used in "C# Visual Studio" and this could be used in two ways:

- Desktop application
- Window application

We use local/internal database to stores data and record of laptop seller and Customer. Desktop application could be used without internet and user could be satisfied with this application and it's useful for every user. Window application could be used with internet and is useful for online projects and for offline applications.

2.2 Application Overview

The main objective of the project is laptop sales and purchase. This "Laptop Management System" consist only on admin site. User can visit shop and buy product as they required. Only admin interact with system by authorization. Whereas admin can add product, delete product update stock and price etc.

- Admin must login for interact with system.
- Admin can any modification after authentication.
- If customer want to buy anything then he/she must visit shop and purchase product as he/she required.
- All required record of customer will save in database.
- Customer can order by sending Gmail or contact us with our given cell no etc.

2.3 Design Languages

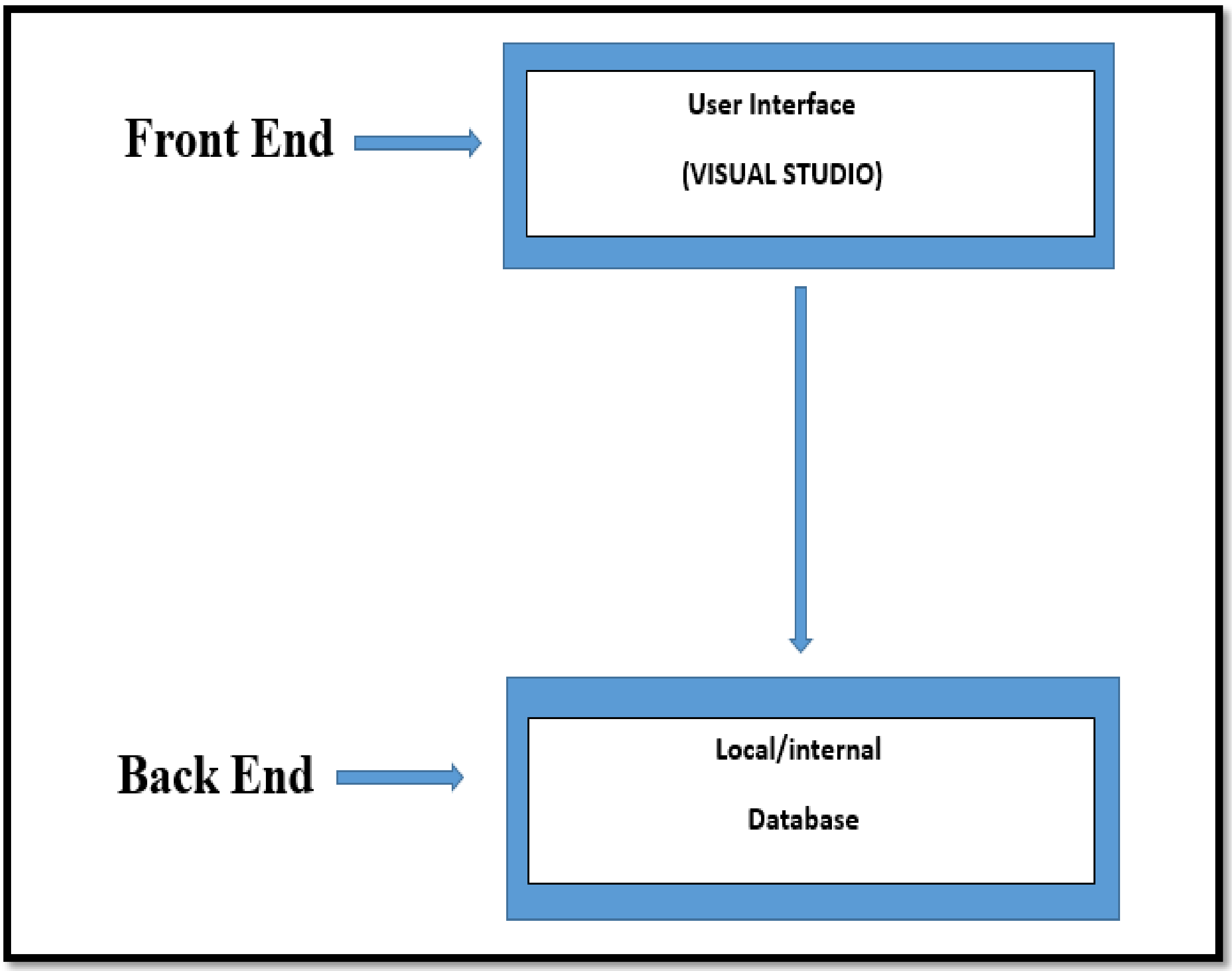
- C-Sharp(C#) for UI/UX.
- SQL for databases.

CHAPTER NO 3:

SYSTEM ARCHITECTURE

3.1 Architectural Design

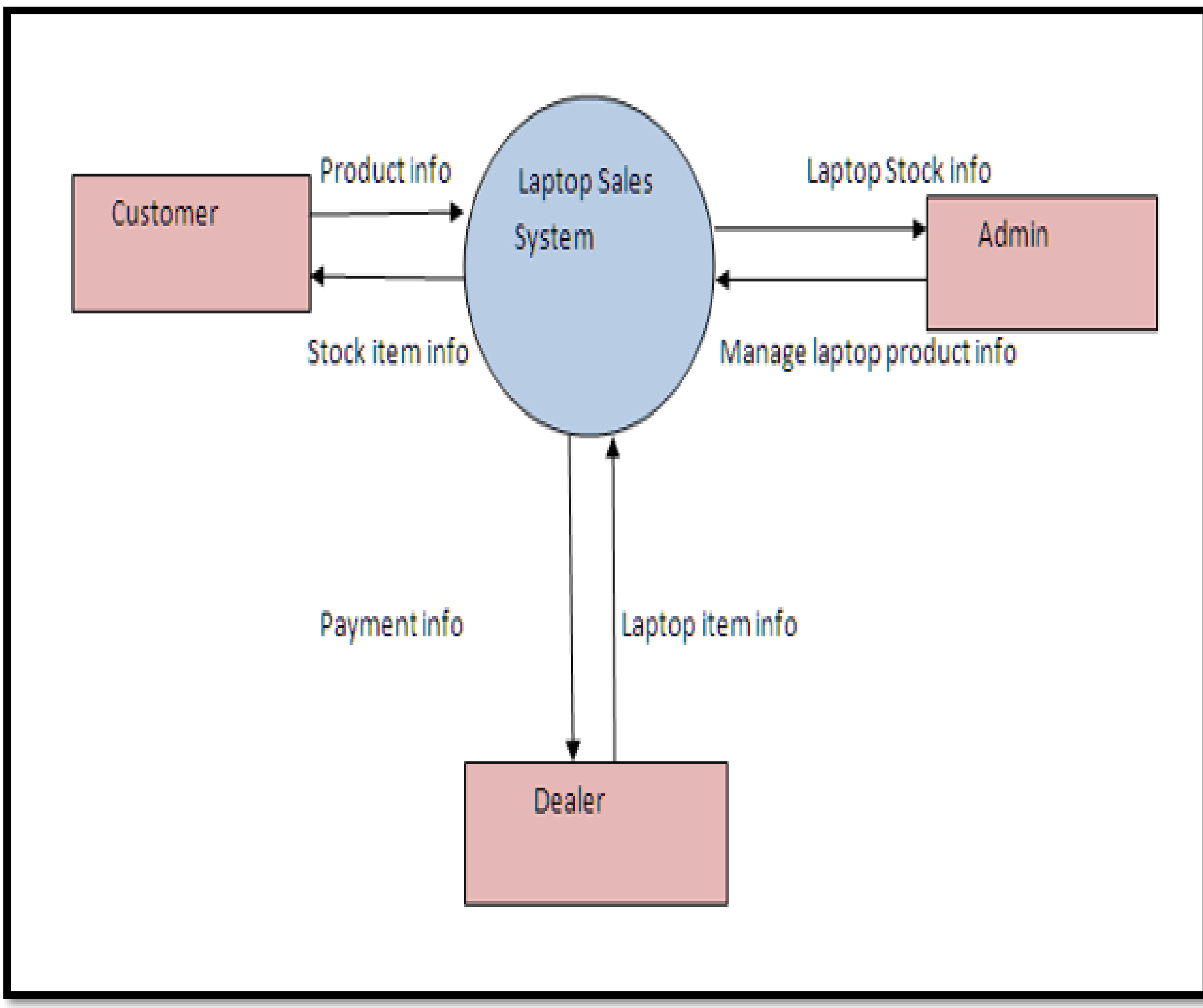
The interface of the system will design in C-Sharp language by using tool **VISUAL STUDIO.** This application will run on a single personnel PC in a shop because it developed as a desktop application. All of the records and entries are stored in local/internal/offline database.



3.2 Decomposition Description

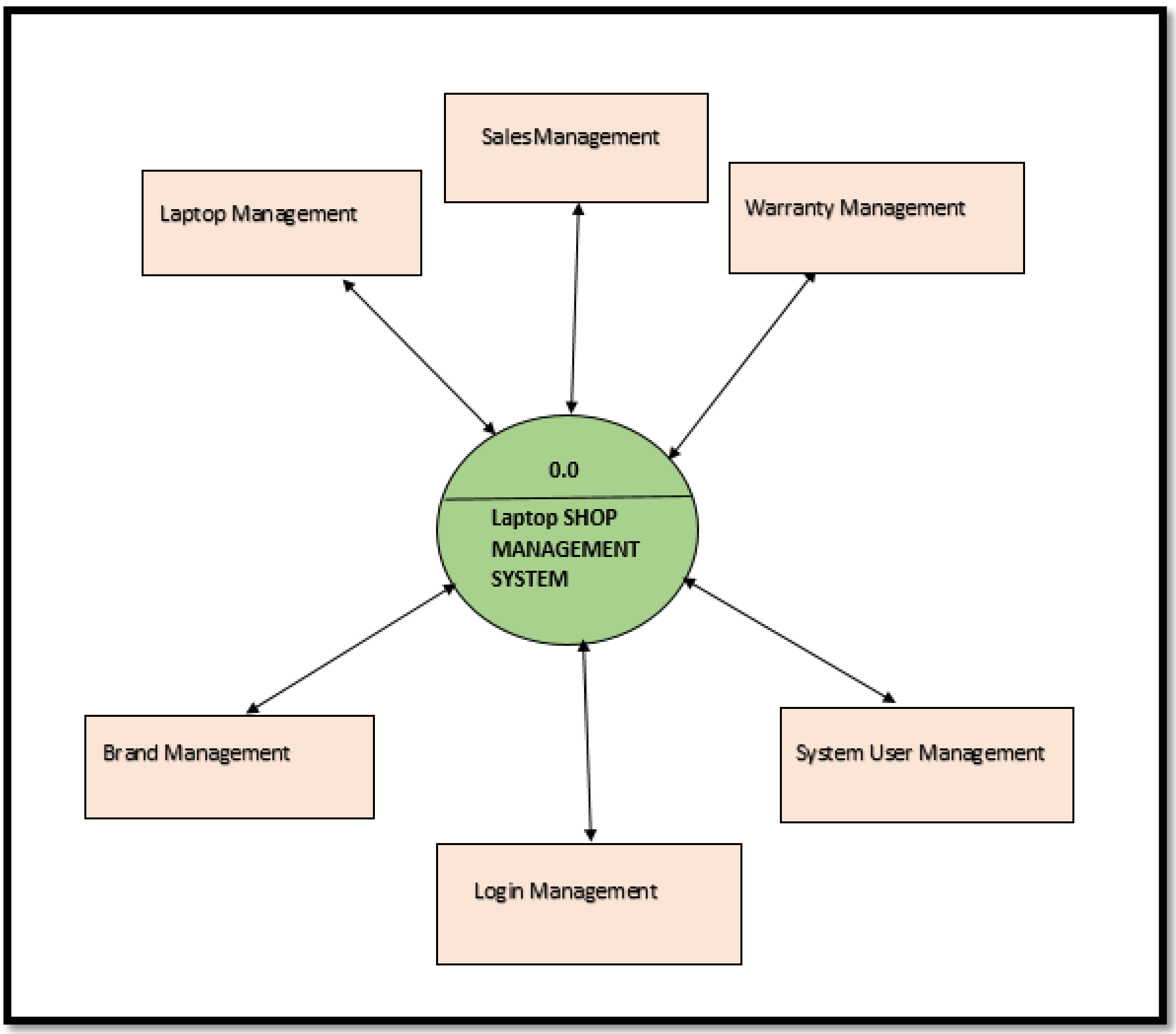
- ### Context Level

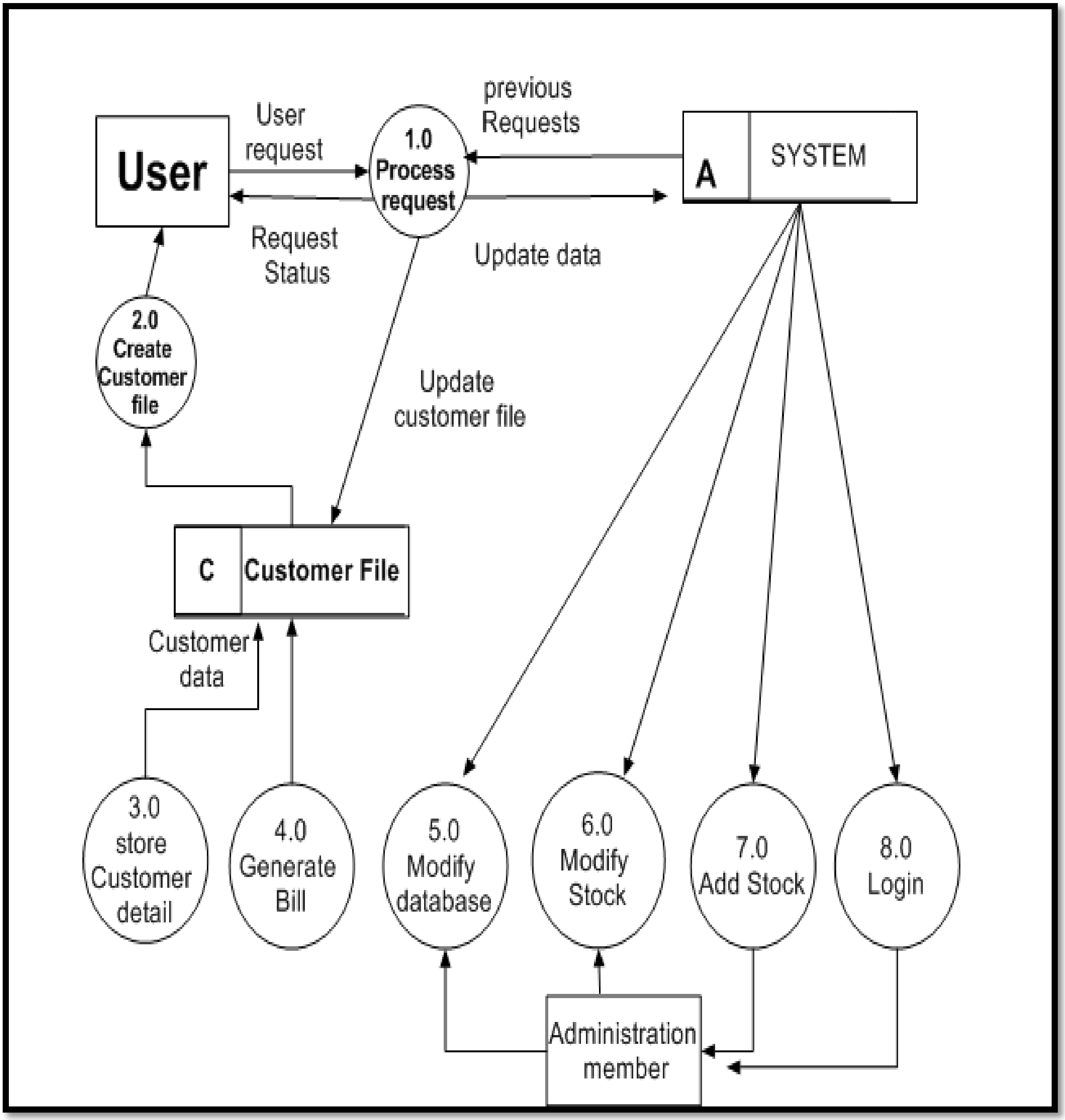
  - Context level is a top level data flow diagram. It only contains one process node that generalizes the function of the entire system.
  - A context level diagram is drawn in order to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.in external entities.
  - It defines the boundary between the system and the part of the system.
  - It exchanges data with outside entities. They are often used for high level planning.



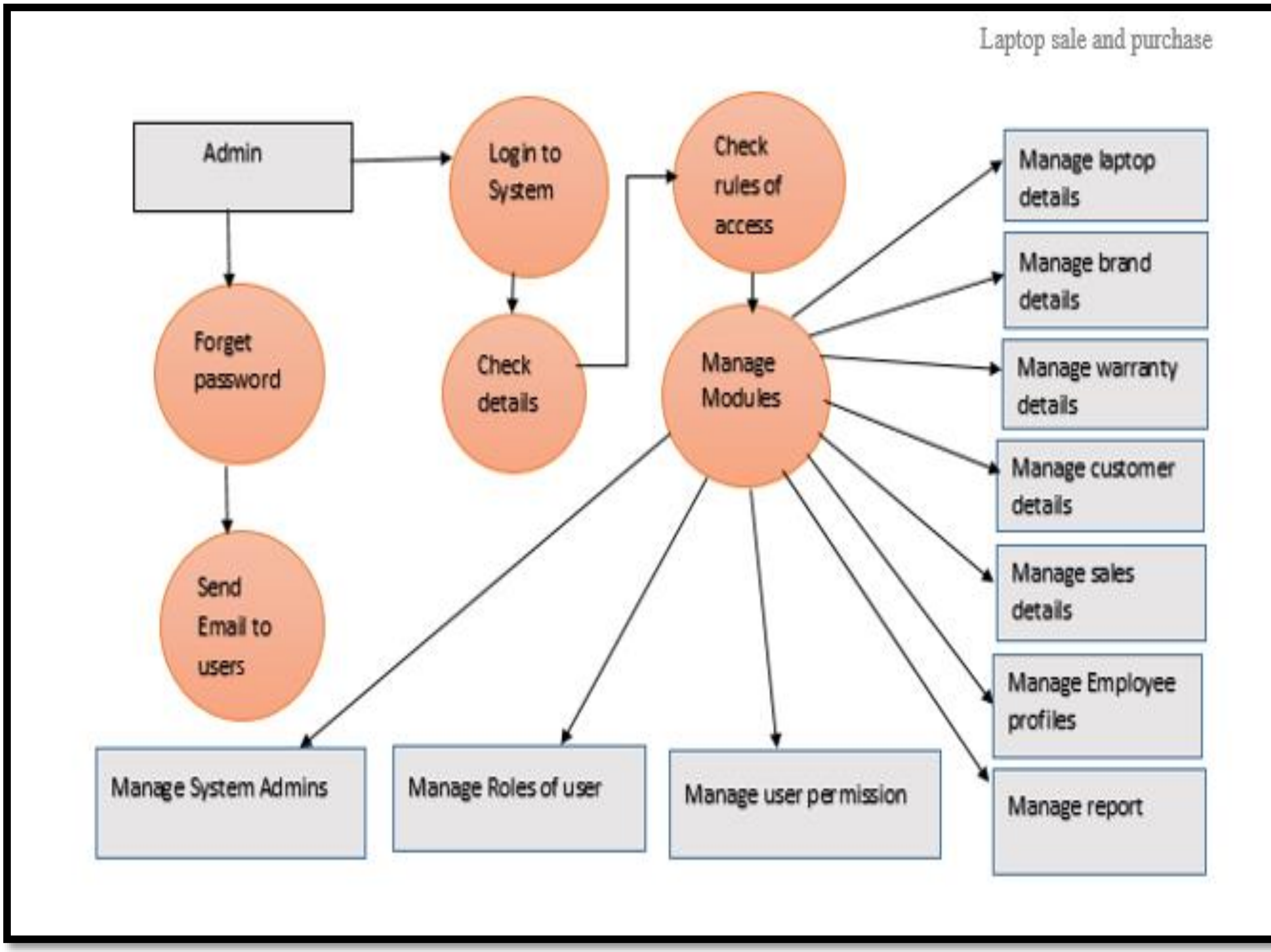
- ## DFD (0-level)

  - (0-level) data flow diagram may only contains one process node (process-0) that generalizes the function of the entire system in relationship to external entities.
  - A (0- level) diagram is drawn in order to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.
  - Level-0 context diagrams shows a system as a single process with inputs and outputs flowing to or from external entities.




User
User
request
1.0
Process
request
previous
Requests
A
SYSTEM
Request
Status
Update data
2.0
Create
Customer
file
Update
customer file
C
Customer File
Customer
data
3.0
store
Customer
detail
4.0
Generate
Bill
5.0
Modify
database
6.0
Modify
Stock
7.0
Add Stock
8.0
Login
Administration
member

- **Detailed Diagram**

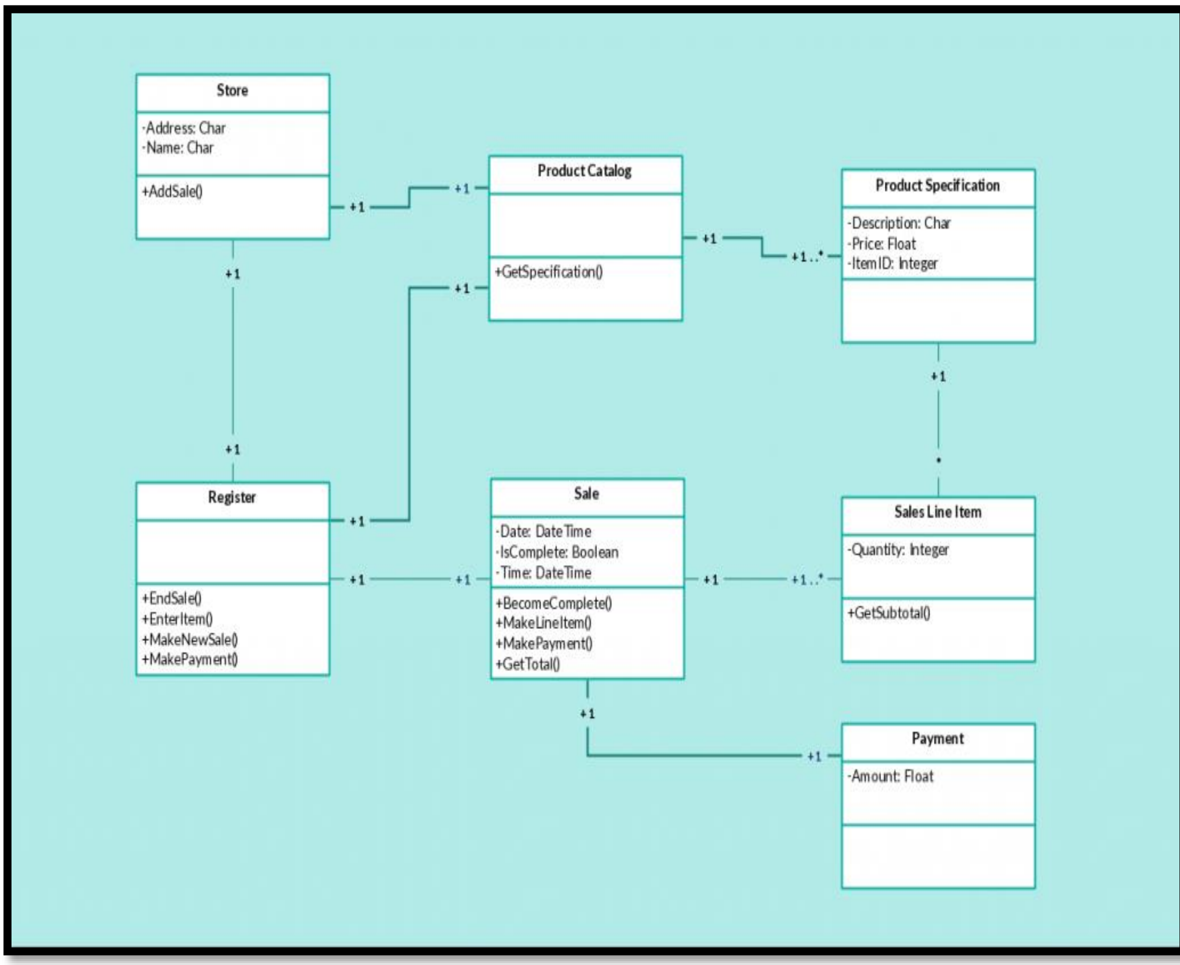


- **Class Diagram**

The static diagram which represents the static view of an application. It is known as a "**Class Diagram**" A class's attributes, operations, and the system's constraints are described by the class diagram. Due to their ability to be mapped directly with object-oriented languages, it is used for modeling in such systems. Also known as a structural diagram, it is a collection of constraints, associations. In a class diagram, it is necessary that there exists a relationship between the classes. The similarity of various relationships often makes it difficult to understand it. Below are the relationships which exist in a class diagrams.

It may consist of advantages of Class Diagrams:

- Any simple or complex data model could be illustrated using the class diagram to get maximum information.
- The schematics of an application could be understood with the help of it.
- Any requirement to implement a specific code could be highlighted through charts and programmed to the described structure.
- A description which is implementation independent could be provided and passed on to the components.
- The Class diagram below shows a point of sales system. Each Class represents different stages involved in a retail transaction



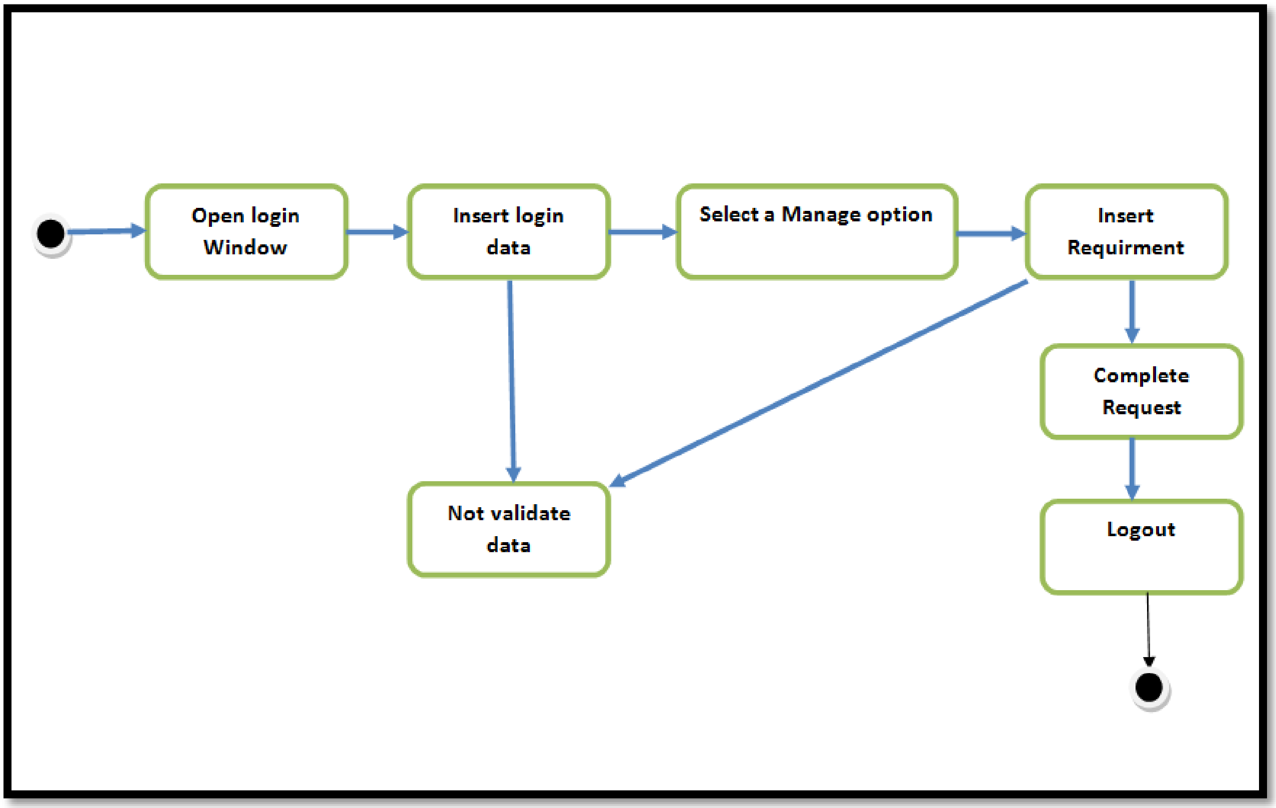
- **Activity Diagram**

  - Activity diagrams are typically used for business process modeling, in this modeling the logic captured by a single use case or usage diagram, for modeling the detailed logic of a business rule.
  - Although, UML activity diagrams could model the internal logic of the entire complex operation.
  - It would be better to simply rewrite the operation so that it is simple enough that you don’t require an activity diagram in many ways in it.
  - UML activity diagrams are the object-oriented equivalent of flow charts and data flow diagrams from structured development of the system.
  - UML activity diagrams refer to software engineering method that is used to describe work flows, business processes and other procedures
  - Activity Diagrams describe the workflow behavior of the current system
  - Activity diagrams are used in process data modeling and analysis using basic requirement engineering.
  - A typical business process which analyze several external incoming events that can be represented by activity diagrams.

They are most useful for understanding work flow analysis of synchronous in it. Activity diagrams are used for

- Documenting existing process
- Analyzing new process Concepts
- Finding reengineering opportunities

- ## Activity Diagram



3.3 Design Rationale

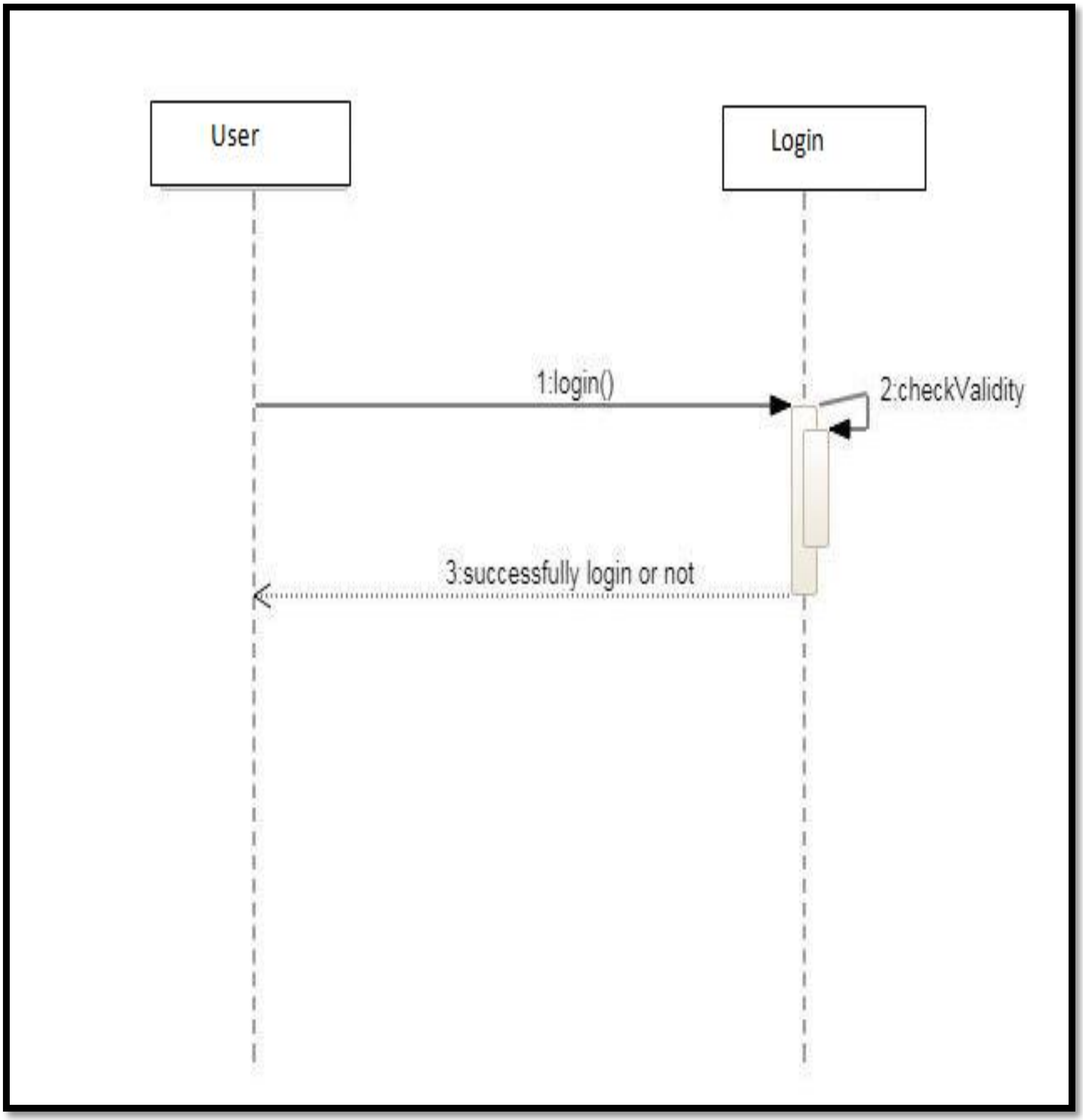
This application **"Laptop Sales and Purchase**" is a desktop application which will be developed/coded in C-Sharp1(c#) language and the tool that is used for this language is **VISUAL STUDIO.** This tool is easy for developed this application. To records or save data we use the internal/local/offline database by connecting.so, the storing activity performed by the local database.

- ## Sequence Diagram

Sequence Diagrams are interaction diagrams that detail how operations are carried out

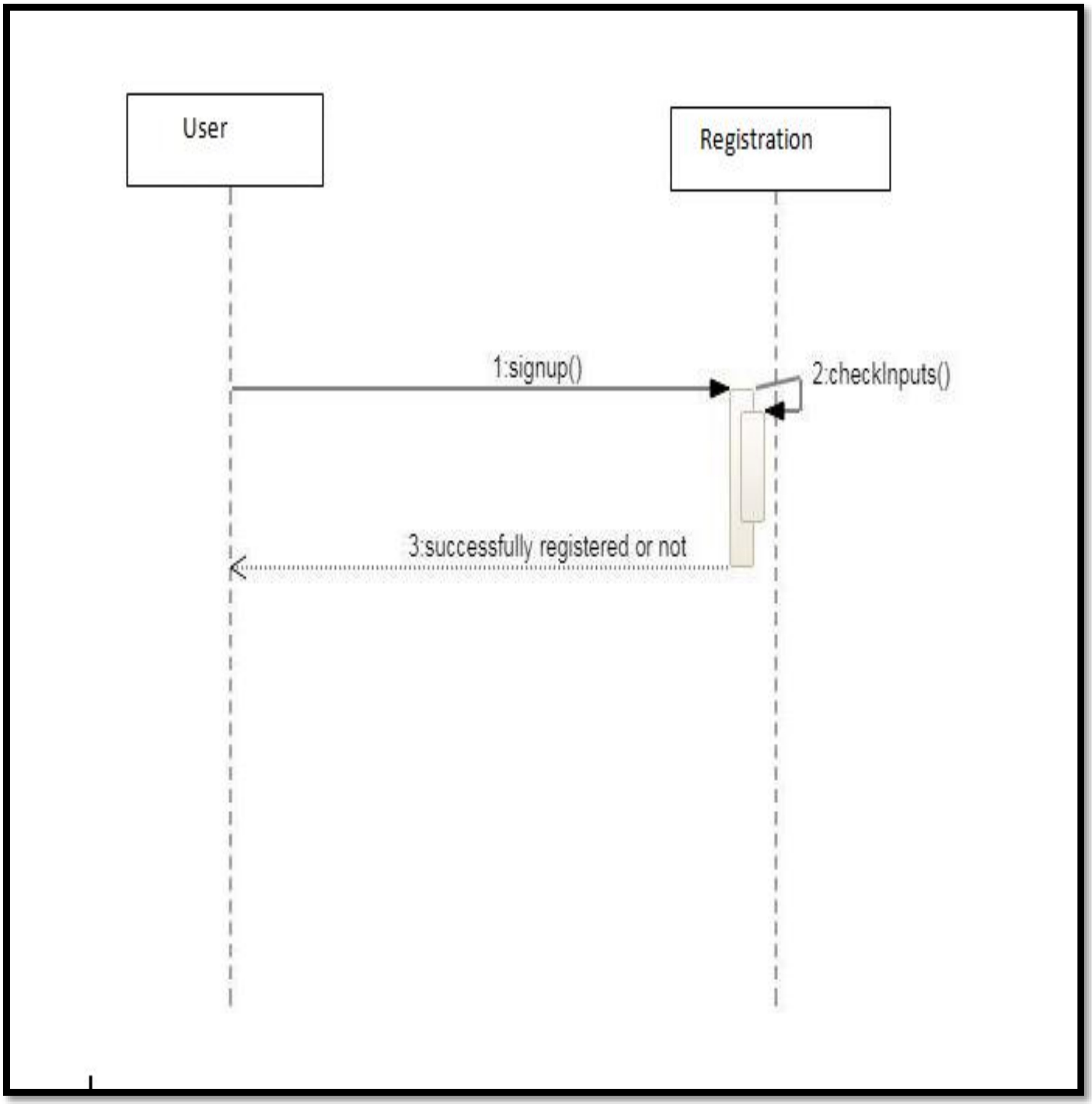
- Interaction diagrams model important runtime interactions between the parts that make up the system.
- A sequence diagram is made up of a collection of participants.
- Participants – the system parts that interact each other during the sequence.
- Classes or Objects – each class in the interaction is represented by its named icon along the top of the diagram.

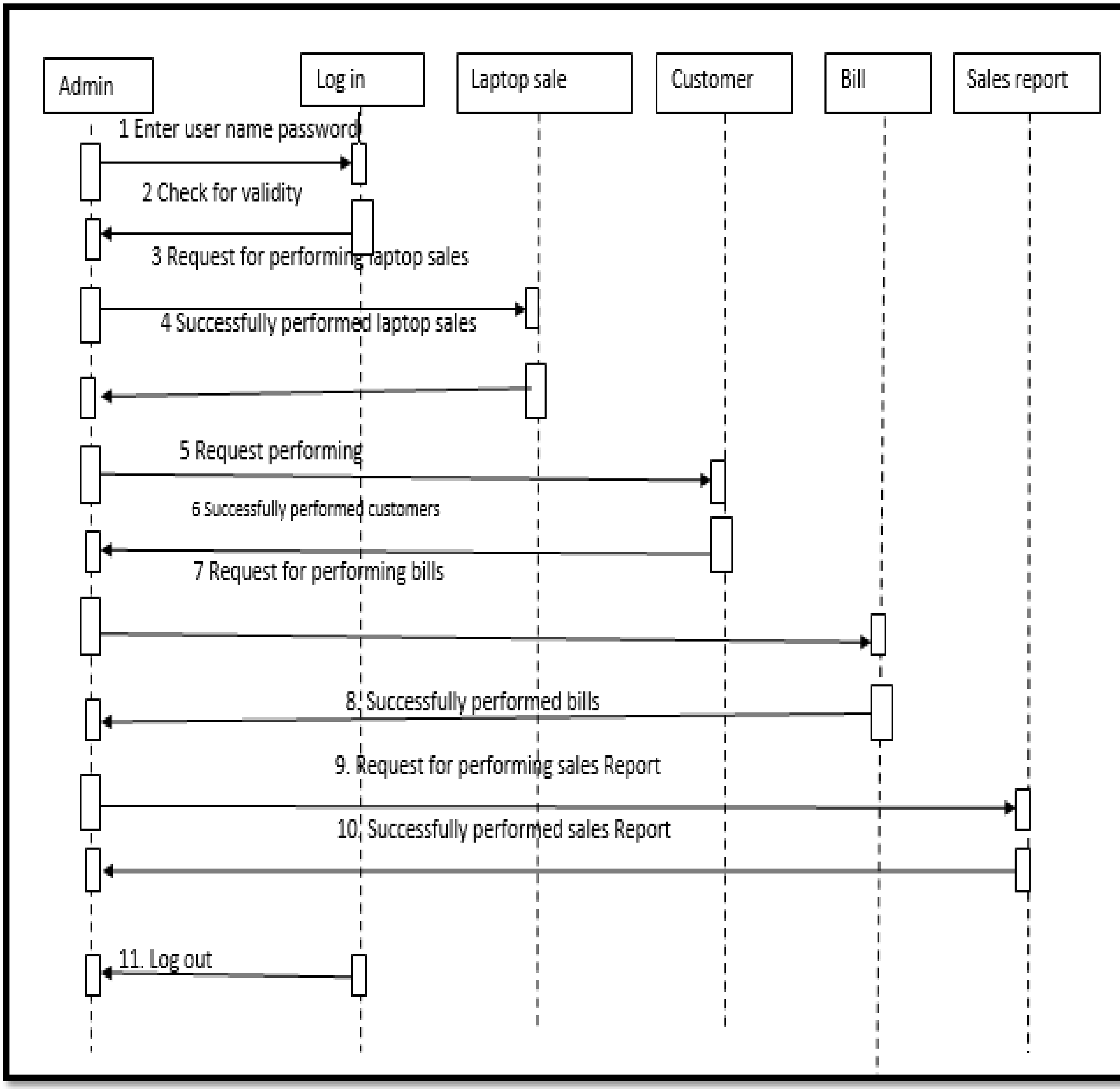
- **Login**



User
Login
1:login()
2:checkValidity
3:successfully login or not

- **Signup**


User
Registration
1:signup()
2:checkInputs()
3:successfully registered or not


Admin
Log in
Laptop sale
Customer
Bill
Sales report
1 Enter user name password
2 Check for validity
3 Request for performing laptop sales
4 Successfully performed laptop sales
5 Request performing
6 Successfully performed customers
7 Request for performing bills
8. Successfully performed bills
9. Request for performing sales Report
10. Successfully performed sales Report
11. Log out

- ## Collaboration Diagram

Collaboration diagrams illustrate interactions between objects the collaboration diagram illustrates messages being sent between classes and objects (instances). Collaboration diagrams express both the context of a group of objects (through objects and links) and the interaction between these objects (by representing message broadcasts) they are very useful for visualizing the relationship between objects collaborating to perform a particular task

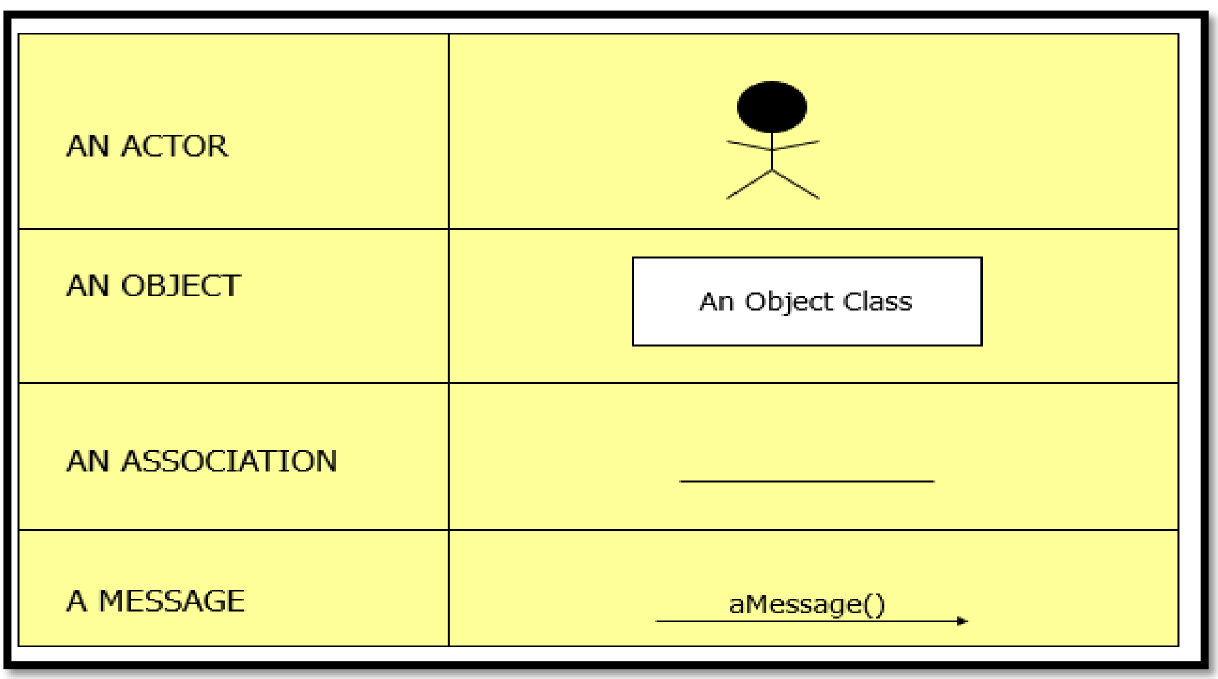
It provide a good interaction between objects which may be difficult to see at the class level
There are three primary elements of a collaboration diagram:

- Objects
- Links
- Messages

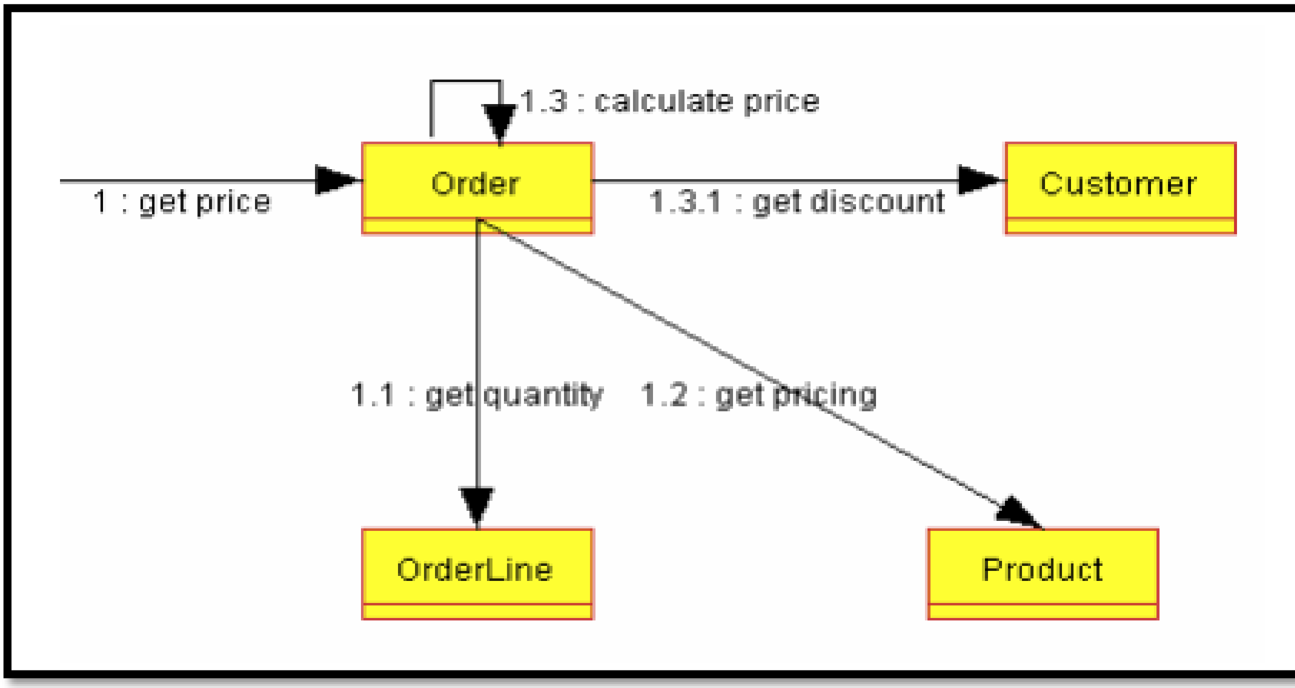
So we can create a collaboration diagram by:

- Create links between the objects
- Create messages associated with each link
- Add sequence numbers to each message corresponding to the time-ordering of messages in the collaboration

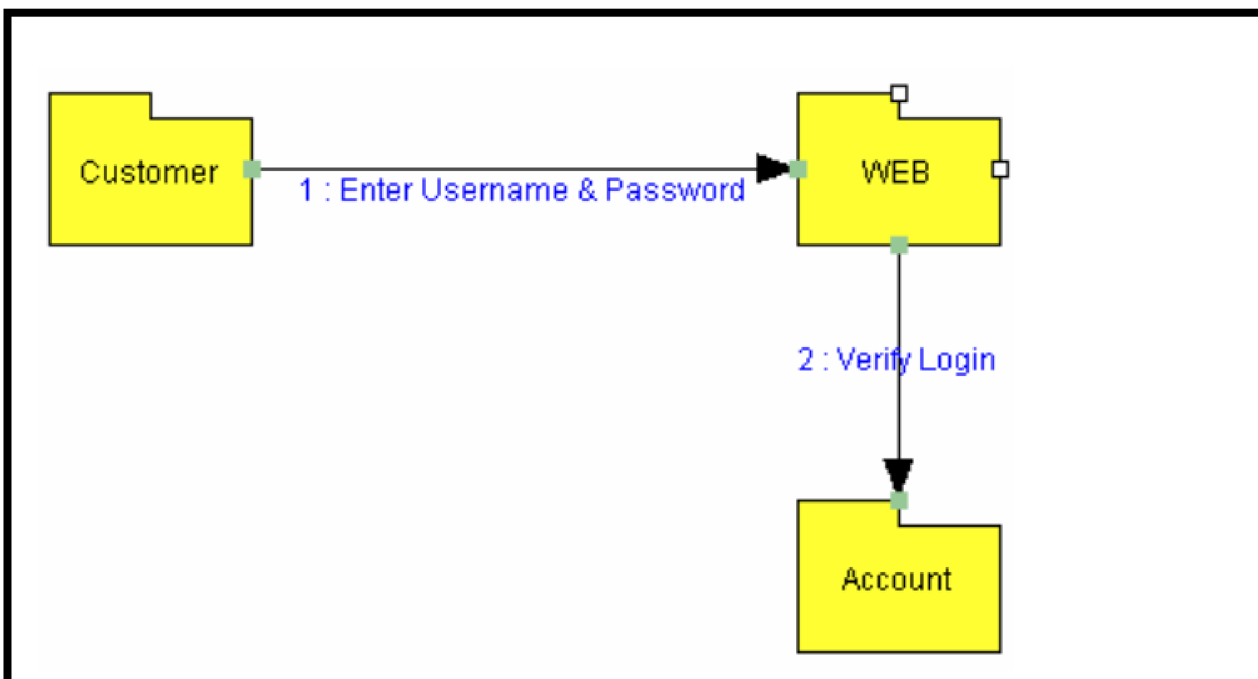
- ## Collaboration Diagram Syntax



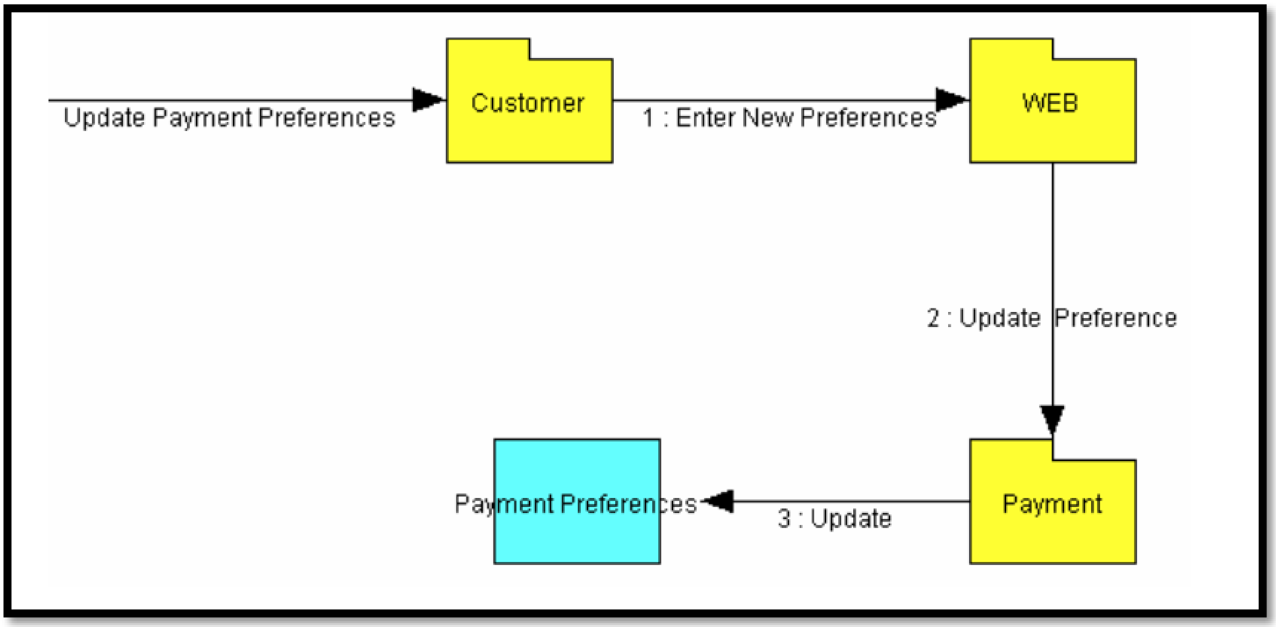
- **Collaboration Diagram**



- **Collaboration Login**



- **Collaboration Payment and Updation**



CHAPTER NO 4:

DATA DESIGN

4.1 Data Description

- Different constraints/checks are applied in it.
- Required data of customers as condition stored in the database. All of this data will be accurate which stored in database.
- We choose C-Sharp**(C#)** language for developing or designing this application.
- **VISUAL STUDIO** tool is used for this language.
- **Laptop Sales and purchase** is a desktop application and runs on a single PC at shop.

4.2 Data Dictionary

Explanation of tables are as follows:

- **Data Dictionary of Admin Login Table:**
    - Username
    - nvarchar(50)
    - 256 characters
    - A to Z and a to z

    - Password
    - Admin-password
    - int
    - 0-9

    - E-mail
    - Email address
    - nvarchar(50)
    - 256 characters
    - 0-9, a to z, A to Z, and special characters.

Table:

Name	Data Type	Allow Nulls	Default
Username	nvarchar(50)	☐	
Password	int	☐	
email	nvarchar(50)	☐	
		☐	

Query of this table:

```
CREATE TABLE [dbo].[AdminLogIn] (
    [Username] NVARCHAR (50) NOT NULL,
    [Password] INT           NOT NULL,
    [email]    NVARCHAR (50) NOT NULL,
    CONSTRAINT [PK_Table] PRIMARY KEY CLUSTERED ([email] ASC)
);
```

Database of Login table:

Username	Password	email
NULL	NULL	NULL

- **Data Dictionary of Laptop Record Table:**
    - Customer name
    - nvarchar(50)
    - 256 characters
    - A to Z and a to z

    - Customer ID
    - User-Id
    - Int
    - 0 to 9

    - Laptop Model
    - Model no
    - int
    - 0 to 9

    - Laptop Price
    - Price
    - int
    - 0 to 9

    - Category of Laptop
    - Category
    - nvarchar(50)
    - 0 to 9, a to z, A to Z.

    - Laptop color
    - Color
    - nvarchar(50)
    - 0 to 9, a to z, A to Z.

Table:

	Name	Data Type	Allow Nulls	Default
	Customer Name	nvarchar(50)	☐	
🔑	Customer ID	int	☐	
	Model	int	☐	
	price	int	☐	
	Category	nvarchar(50)	☐	
	Colour	nvarchar(50)	☐	

Query of this table:

```
(
    [Customer Name] NVARCHAR(50) NOT NULL ,
    [Customer ID] INT NOT NULL,
    [Model] INT NOT NULL,
    [price] INT NOT NULL,
    [Category] NVARCHAR(50) NOT NULL,
    [Colour] NVARCHAR(50) NOT NULL
)
```

Database of Laptop Record table:

	Customer Name	Customer ID	Model	price	Category	Colour
*	*NULL*	*NULL*	*NULL*	*NULL*	*NULL*	*NULL*

- **Data Dictionary of Stock Updated Table:**
    - Item name
    - nvarchar(50)
    - 256 characters
    - A to Z and a to z

    - Item Number
    - Int
    - 0-9
    - 18 Digits

    - Item ID
    - User-Id
    - Int
    - 0 to 9

    - Item Price
    - int
    - 0 to 9
    - A-Z or a-z

    - Item Quantity
    - int
    - 0-9 and a-z and A-Z and special symbol.

Table:

	Name	Data Type	Allow Nulls	Default
	Item Name	nvarchar(50)	☐	
	Item Number	int	☐	
🔑	Item ID	int	☐	
	Item Price	int	☐	
	Item Quantity	int	☐	
			☐	

Query of this table:

```
CREATE TABLE [dbo].[Stock Update]
(
    [Item Name] NVARCHAR(50) NOT NULL ,
    [Item Number] INT NOT NULL,
    [Item ID] INT NOT NULL,
    [Item Price] INT NOT NULL,
    [Item Quantity] INT NOT NULL,
    CONSTRAINT [PK_Table] PRIMARY KEY ([Item ID])
)
```

Database of Stock update table:

	Item Name	Item Number	Item ID	Item Price	Item Quantity
	NULL	NULL	NULL	NULL	NULL

- **Data Dictionary of Bill table:**
  - Date
  - Date
  - Shopping date
  - Varchar(50)
  - 256 Characters
  - 0-99999999999999999999 and special symbol

  - Customer Name
  - name
  - Name Of customer
  - Varchar
  - 256 Characters
  - a-z, A-Z

  - Total Bill
  - Bill
  - Total bill of customer
  - Integers
  - 18 digits
  - 0-9

Table:

	Name	Data Type	Allow Nulls	Default
🔑	Date	date	☐	
	Customer Name	nvarchar(MAX)	☐	
	Total Bill	int	☐	
			☐	

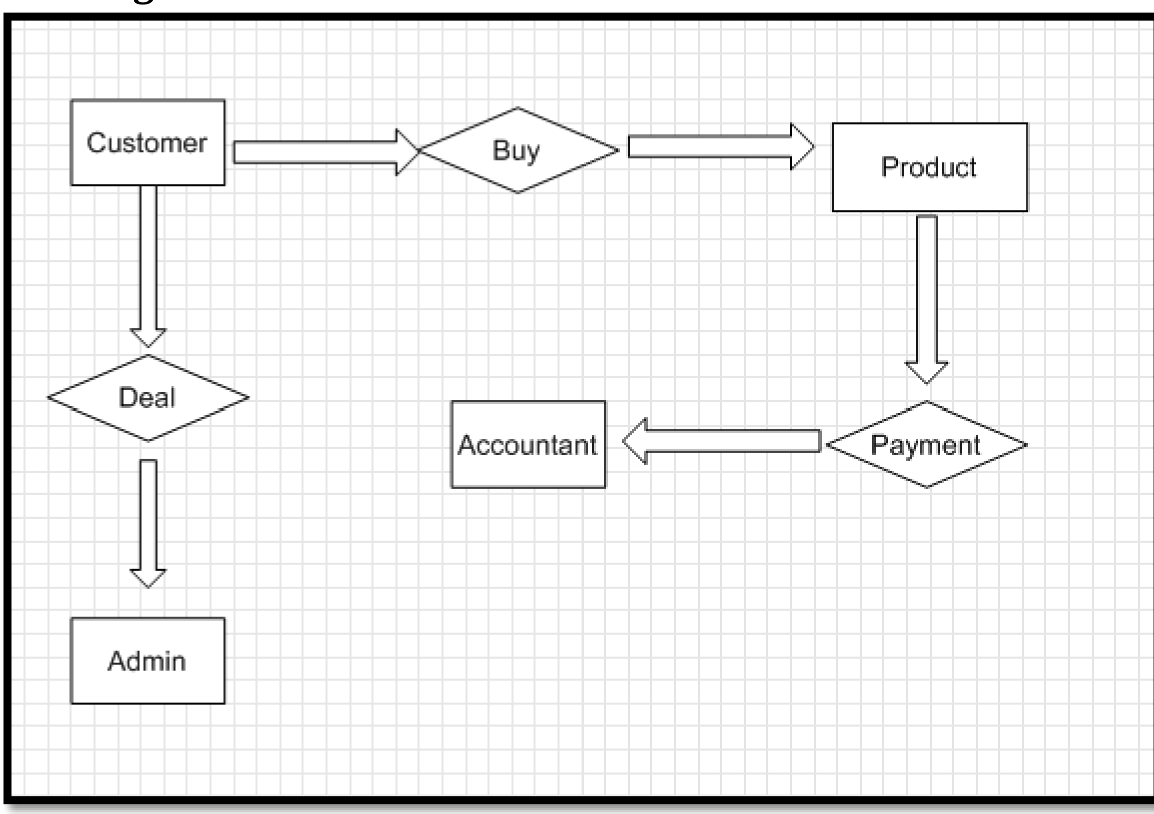
Query of this table:

```
CREATE TABLE [dbo].[Table]
(
    [Date] DATE NOT NULL PRIMARY KEY,
    [Customer Name] NVARCHAR(MAX) NOT NULL,
    [Total Bill] INT NOT NULL
)
```

Database of Bill table:

	Date	Customer Name	Total Bill
*	*NULL*	*NULL*	*NULL*

- **ER-Diagram**

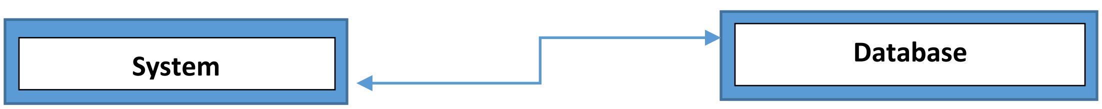


CHAPTER NO 5:

COMPONENT DESIGN

5.1 Component Design

- The system is capable to do work according to the direction. Every simplest or complex
- Component is responsible to do work which is assigned to it. The system considers every
- Component as best efficient way component for the mutual cooperation of the whole
- Process. Component design follows as:-



Front end **Back end**

The short detail of the component design working describes as the admin after login to the system will choose an option for the submission or other action. Consider an example if admin want to generate bill then admin will press the button of total bill. The given data in the text box will be stored in the database, similarly admin can do updating by the corresponding button in it.

CHAPTER NO 6:

HUMAN INTERFACE DESIGN

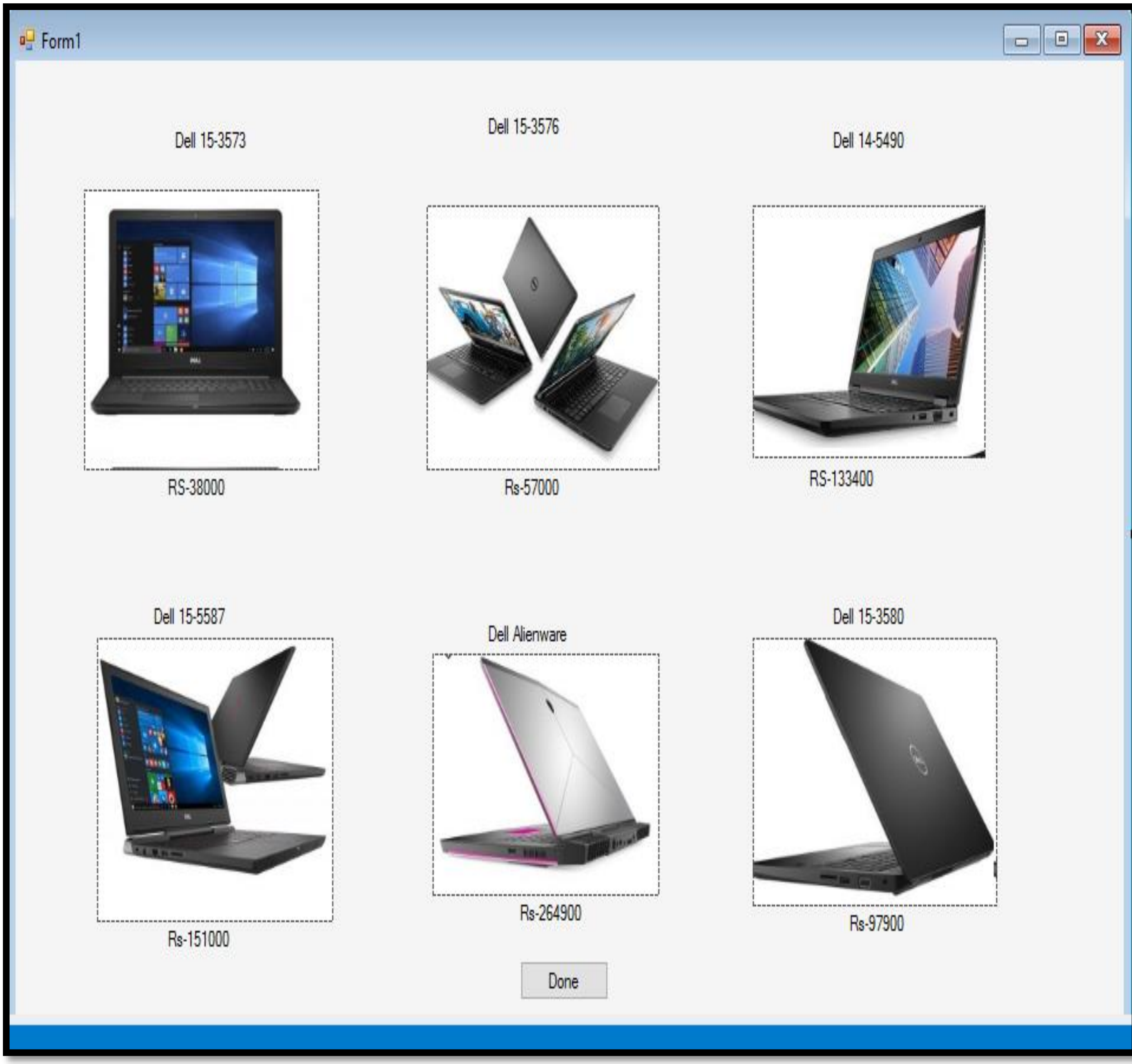
6.1 Overview of User Interface

This section describe about the functionality of the different forms.

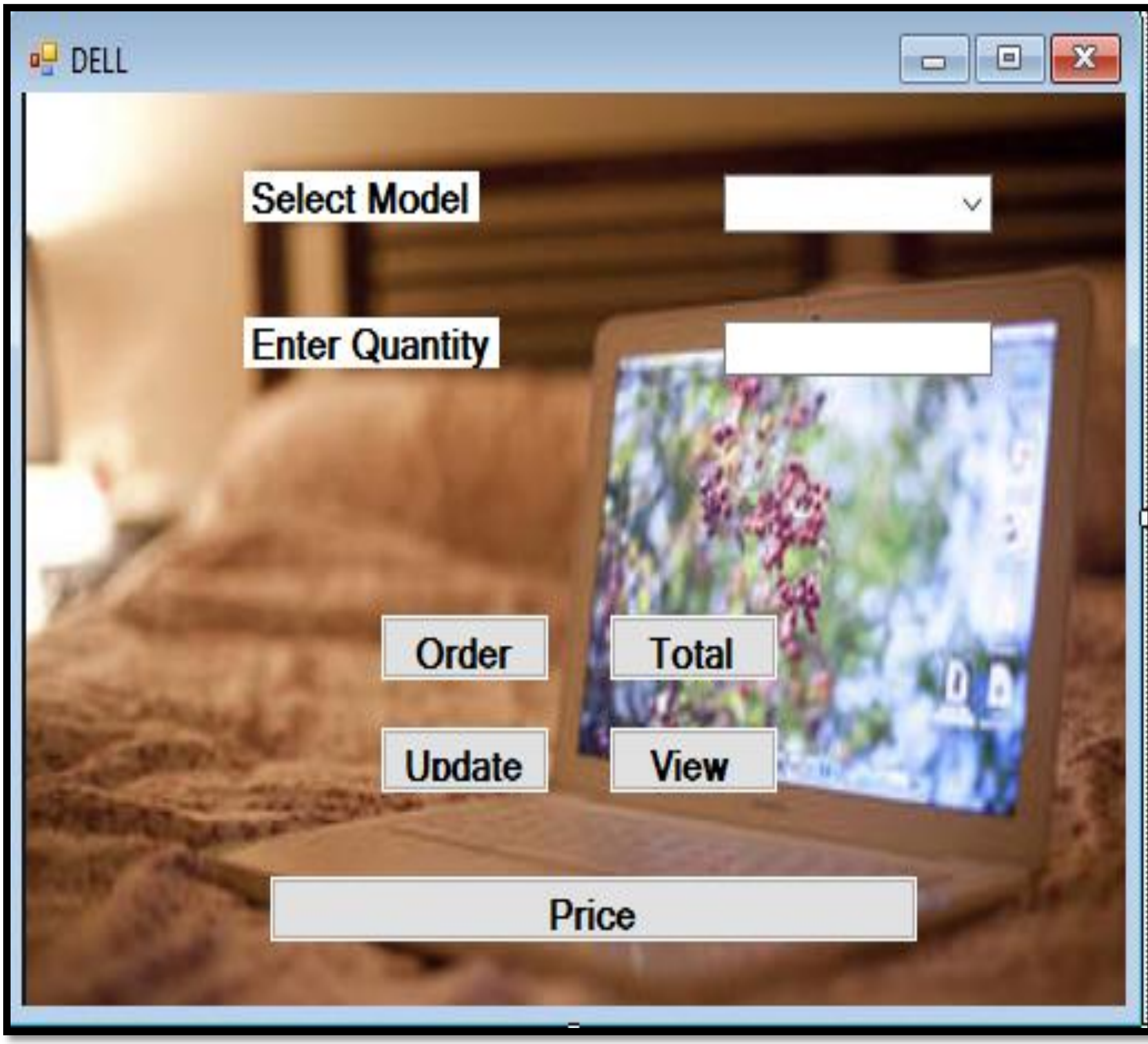
6.2 Screen Images

Login Form

Company Selection Form

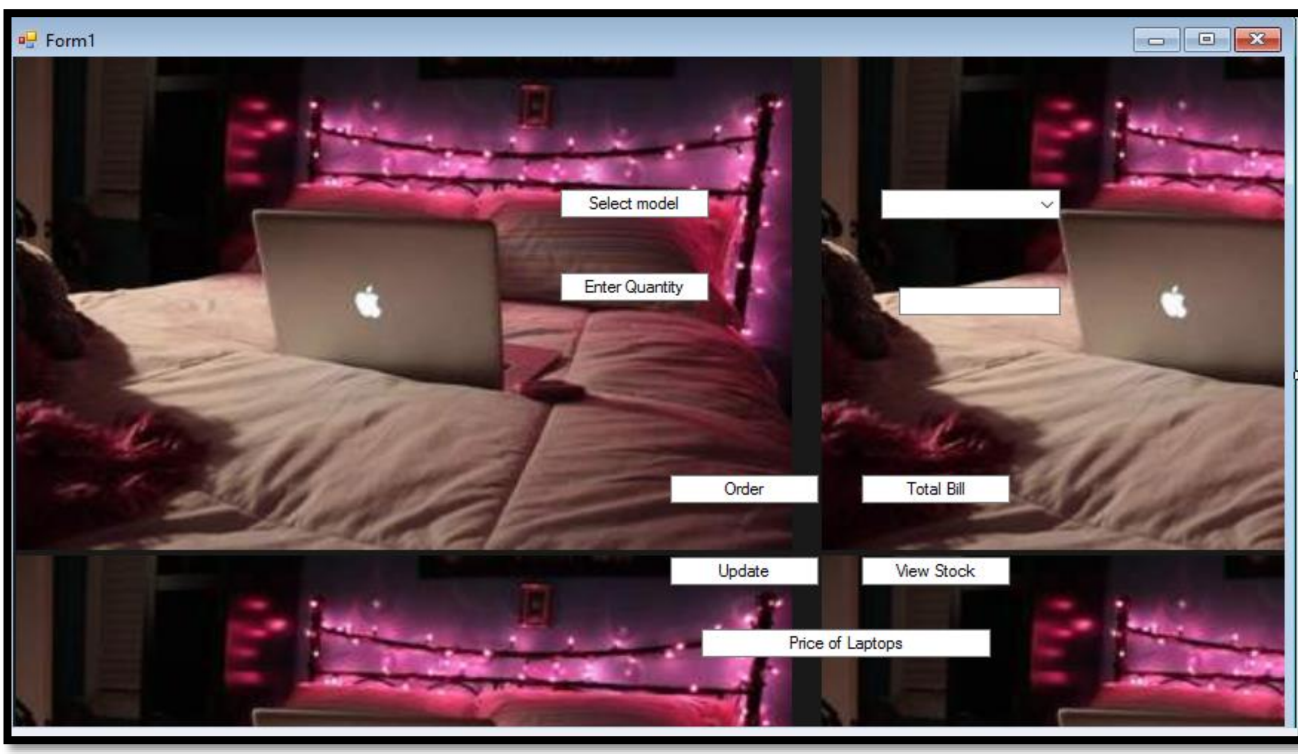


Dell Laptops with prices



Main form where all laptops model shown

Dell Updation Form

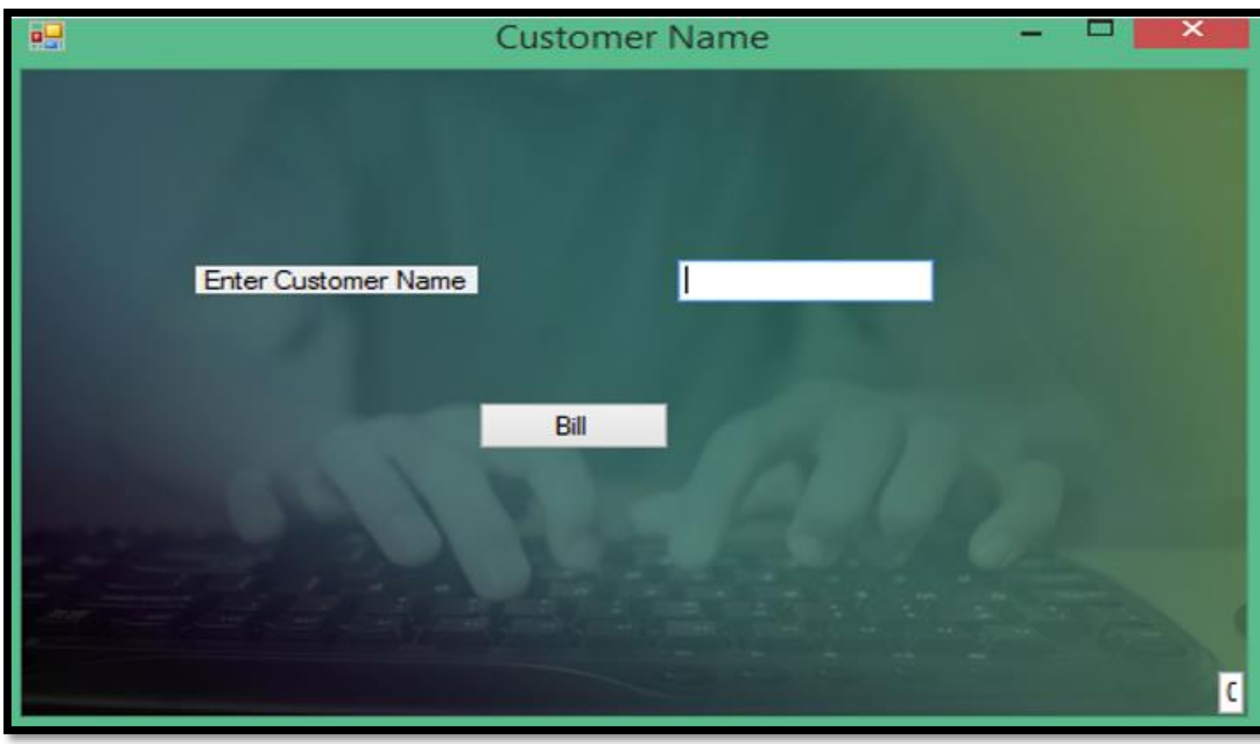


Apple Main form

Apple laptops with prices

Apple Updation Form

Bill Generation Form



Customer information Form

HP Main Form

HP Laptops with Prices

HP Updation Form

6.3 Screen Objects and Actions

- **Login**

  Login form is used for admin. It will be used for every time when system will shut down and restart this application. This form is necessary for interact with system.

- **Main/Selection Form:**

  After admin login in this application then this form will be open .This form has a different mobile companies. The admin select the company that customer want to purchase a mobile phones. May be another companies will be mentioned in this form.

- **Customer Info Form:**

  After the selecting a mobile from shop customer record save in local/internal database by using this form. This form has a different customer information entries like as name, cell no etc.

CHAPTER NO 7:

REQUIREMENT MATRIX

Components: Requirements from SDD (Use case):	Login logout Component	Main Functionality Component	Settings Component	Social Component	Trade Component	Data Services Component	Remote Server Component
UC1	x	x					
UC2	x	x					
UC3	x	x					
UC4	x	x	X				
UC5	x		X				x
UC6		x					
UC7	x	x					
UC8							